Today we are going to be discussing EPA 400. It is very important in terms of emergency response health physics because it sets guidelines for radiation doses for the public during an emergency and mentions at least to some point food and ingestion pathway limits as well. So I actually did put EPA 400, the latest version, on this week's module. If you would like to read some of the more interesting chapters that their quick chapters are three, four, and five, and it talks about Pags and how they came to be and what goes on with them. There's also another paper I'm going to put out that I would like you to look at which talks about some evacuations that happened during Fukushima and it's an article that discusses doing planned evacuations instead of doing something fly by night because it talks about how many people died in Japan due to poorly planned evacuations. I think in this article it discusses how they moved a nursing home, and they kept them in a bus for over 48 hours without food or water, and like six or seven of them died because of dehydration. So, yeah, it's, yeah, so the joke, well, not the joke, but the sad thing is that if they're in a nursing home, they're probably in their advanced years, their idea of the possibility of them getting a cancer due to the radiation from Fukushima, non-existent. So you evacuate all those people for really no reason whatsoever. But anyway, some really good stuff to read. I would highly recommend all of those chapters as well as the article, because we'll probably have at least one little question on a quiz for next week's article. So what I would like to do is, let's start off at least with some definitions. So one definition that you're going to hear a lot is something called a PAG. It is a protective, protective action guide. I mean, in all purposes, it's a dose guideline when protective actions are warranted. And these are usually in, so this is like dose, how many, you know, Miller-in. We'll talk about specific dose limits here in the secondary guidelines. And again, notice, in radiation, emergency response, we are not talking about laws. There are very few laws about when you should evacuate, when you should shelter in place, when you should move people out. So there's very few laws about that. There's only guidelines out there. So the guidelines give you room to actually think about what you're doing. So should I really, it says I should evacuate? weight at one rim, but it's only a guideline. Maybe I should shelter and place at one rim. Maybe people shouldn't move out if they're only going to get two mill rim in the first year. But anyways, this is a dose limit, something accumulated over time, and this is usually projected. So it's a protective action guide. You actually sometimes will hear the word PAD, which is protective action decision.

So what have you done? What is the decision by the public, the decision maker to help protect the public, and it's usually based on a protective action guide. You'll hear another one, and we'll start talking about this more when we do the Fermat manual, a DRL. It's a derived response limit. So this is a measurable value that is projected to meet a PAG. So in this case, it might be, let's say, how many miller rim per hour you're measuring. So this would be something where we would send monitoring teams out and we would tell them, hey, we want to know when you get to five miller rim per hour. Because we projected if you are at five miller rim per hour, then over 100 hours, you will meet the evacuation pack. Because you can't say, this is definitely one rim. People should evacuate, but I can't at least go out and say, this is one milligram, and over 100 hours, it will eventually reach some protective action. So it's called a derived response level. For food, you have something called the derived intervention level, or the dill, derived intervention level. So a measurable amount in food slash liquid that if ingested could meet a path. So you might be monitoring vegetables or truckloads of fruit or something that's coming out of a potentially contaminated area. So if you measured said fruit and it hit, you know, 300 microcure or microar per hour, then there are calculations that have been made that would suggest that, oh, if somebody actually eats this, they would actually reach a protective action guide. So this food should be embargoed. So that's why I keep thinking of it as a derived intervention level, because at this point, you're intervening between food, getting to people. Yes. Isn't it a derived concentration? Yeah. Derive concentration. Okay. Well, the question that Jordan's asking for Internet Landis, she's asking about a DCRL. Off the top of my head, I can't think what it is. I think it's an acronym that's used in the Fermak manual where it's talking about derived concentration

response level or something like that. Okay. So it does it be the same thing. It might be, it might be, It's probably essentially the same thing, but applied to something else. So what has happened, though, in EPA land is EPA protects the environment. And as per, I think it was FERPA, they were set out, or the administrator for EPA was told, hey, thou shalt create guidelines for radiological emergencies. The very, very first one was what they used to call EPA 1992, or it was also EPA 400. it was the same thing. And so when I started out all the way until when I ended my work at the Department of Energy, this was the document that you used for over 20 years. It was what you used for protective action guides following a radiological incident. Then 2013, they actually put out a new one. And the newest, I think they put out in 2017. So this is the current volume. volume.

And it's kind of funny, you think about it, 1992, no changes. And then the last decade, we've had two changes.

So what is interesting, we'll go over about why they made some big changes here in a second. But one thing to point out is in 1992, when they were calculating doses and they were trying to figure out, you know, make calculations about health issues that might occur due to a radiological incident, they were using ICRP, 2630 for dose projections.

Now they're using ICRP 60 for their dose projections.

But that can change. And we'll talk about what flexibility has been put into these documents to make that change. So differences where they talk, differences that exist in this document now is the organizational structures change a little bit. They've kept what they called the three phases. It was like the early, the intermediate, and the late phase. They put new protective action guides, but before EPA relied on their own expertise to develop the guidelines for doses. In the newest documents, they've used all these organizations and inputs for all these organizations to justify their protective action guides. And we'll talk why this is important. And why they could do this, I just put this down here is 44 CFR 351 and Executive Order 12656, give EPA the mandate to produce a protective action guides document. So the official federal document for protective action guides, again, is EPA 400. And that is what we technically base everything off of during an accident. So what do you think change, though, between 1992? and 2000 to justify, besides new science behind, what do you think the only thing they were interested in in terms of radiological releases in the 1990, in the 1980? What? What type of industrial accidents? Nuclear power plant. That is the only thing EPA 1992 was based on. They were concerned about nuclear power plant releases. And so all their assumption, fission products, primarily actually primarily fishing products where what was being pushed out so they were very they were really only concerned about that so the new ones has expanded um rad releases and includes not only nuclear power plants and pps but fuel cycle facilities weapons facilities i'm the same fact for facilities transportation accidents radio pharmaceuticals and launch vehicles radiation dispersal devices, RDDs, and improvised nuclear devices. So now we're not just looking at fission products. We're looking at a whole host of different types of radio nuclei that could be released during any of these events and gives you an idea of how to calculate doses and why the scientific justification of setting specific protective action guides based on dose. more frequently used or because of being like a fear because there was a terrorist attack in certain places so they just know it could get off the world? So the question is for internet land is is the expansion due to like fear of nuclear terrorism or or frequent use and yes all of the above so they just realized because you have to realize when EPA 400 was originally being written they were still concerned about Chernobyl and they were still concerned about three mile island. And so their main focus was, well, this is what we always have to deal with, or this is what has been dealt with in the past. So we really need to focus on nuclear power plants. But then after several years, 9-11 included, hey, it's not just a one thing fits all. We have responded to transportation accidents. We have responded to releases at fuel facilities. There have been releases due to criticality accidents at weapons facilities or even pure. So all of these things have resulted in releases. We're very

concerned about RDDs and INDs. So we want to include that. And what's nice about only thinking about it from a nuclear power plant standpoint is you're only worried about a certain set of fission products. Now you have to kind of estimate what you're going to be worried about. So Pags, now this is kind of interesting. Pags in this manual and how they were calculated and figured out are implemented using calculations and methods in the Furback manual. This is big because, again, 1992, it was EPA. EPA, hey, we're leading the charge, we're doing this. But when you actually have to change and adjust your methods in an EPA document, that is a big deal. It takes a while to do. The Furback manual itself, though, is not a DOE released document. It is a DOE released document, but it doesn't have to go through a lot of the, it's not a federal document. This is a federal guideline document. The Fermat manual is not. It's just an instructional manual. And it doesn't take a lot. As due science happens, the Fermat manual adjusts. And it adjusts much faster. And it is multi-agency. I'll talk about that when we talk about the Fermac manual later. But so this is telling you that all the reasoning in the current EPA 400 is based on Fermat manual calculations and methods.

And that's kind of big. And I'll explain why in a second because of this statement right here. And I pulled this off of EPA 400 because it is very important. So before what I used to do power plant exercises, we would have calculations in the Fermat manual that would tell us one thing. However, EPA 400 from 1992 would say another thing.

States would tell us, well, we don't care what the Fermat Manual says. We actually have it in our action plans. We will follow EPA 400. Therefore, you will follow EPA 400, even though it was based on old dosymmetry data. So what EPA now is put in their document is this statement here. So EPA anticipates that radiological assessment methods will be periodically updated. Oh, shoot, I just turned it off. Come on, come on back. Go on, come on back to daddy. will be parallel to be updated as improved models and methods become available. Therefore, readers are encouraged to review the current version of the Fermac Assessment Manual to understand the most current default radiological methods for simplicity. Specific organ dose thresholds for evacuation sheltering were removed from the manual. I guess I shouldn't have included that last sentence. But what it's just going here is they are telling you, hey, we recognize EPA 400 is great, but we don't change it very often. So if there is ever an issue, we encourage you to look at the improved models and methods, specifically look at the most current version of the Furmac Manual. The most current version of the Furback Manual right now is 2022, I think, or 2023. I think actually January 2020, of course, it released it and it changed stuff, so I have to change my whole lecture for that. But, you know, that's no either here nor there. But yeah, so this is big, big, big, because now when you go into these places and they say, even in their action plans, we will follow EPA 400, then you can point out, yes, this is good. But notice EPA 400, even in their introduction, says that you can follow the format guidelines. Sometimes, though, they'll still go, hey, nope, we still want to follow this. This is what we're doing. And because you are there as a consultant almost, you are there to support them, not to tell them what to do. you go, yes, we'll do that. And actually in TurboFirmac, which you'll see later, is you can actually go to earlier versions of dose models to calculate whatever you're trying to quickly. So it's kind of neat. So say that your consultants say, like, they just don't like these dogs, don't get along whatsoever, and they technically like fire you and ask, tell you the reasons they just make their own questions? I've actually been in situations where they decide they didn't want a federal person. Because if some federal person decided they were in charge and the state this is going to talk about, we don't want you. I've had, I've been in conversations where the state has hung up the phone where they just didn't want to talk to, talk beyond the bridge line anymore discussing this because they got so good. But it was funny. That was between Vermont and New Hampshire and the fence. There's a couple of power plants up there. And Vermont and New Hampshire started arguing about what they should do. And finally, Robots said, we're done. This is an exercise, man. What are you doing? It was, I still remember looking across the table at my friends going, did that just happen? That's kind of weird. So the take-home is, use the most,

the latest methods. So other differences in the protection action guides, FDA's, so the federal drug administration, oh, food and drug administration, sorry, FDA's 1998 updated for interdiction of food was integrated. So FDA's newest or at least latest ones of food interdiction levels were incorporated. FDA decision to lower the PAG for administration, admin of of stable iodine, so potassium iodide, was lowered to 5-rim. They lowered a protective action guide. They removed intermediate phase, relocation. So there's a protective action that's called relocation, where you move people out for a little bit of time. It used to be 5-rim over 50 years. So if you thought you were, if calculations were done, that suggested that the population might get five rim over a 50-year period, then the protective action guide was to relocate those people for some amount of time, so they would not get that dose. It was finally recognized that trying to really calculate five-rim over a 50-year period is really hard, and there's a lot of conservatism based on the calculations to get there. So EPA removed that level now. So there's no 5-year, 50-year protective action guide. But all other Pags stayed the same. And we're going to go over those in detail. $400 is separated an emergency in three separate phases. The first one is the early phase. And here, protective action guides may be based on default procedures. It can last, so may last hours or just. days. So that's the early phase of an event. And what I mean by based on default procedures is if, let's say you have a site area emergency, the procedures might be to evacuate in all directions two miles.

Because we know the accident is happening. First thing we should need to do is evacuate. We don't care. We're not making any calculations. Default says do this. Get rid of this. Let's do this.

So you put if it is suspected that like I know in San Luis Obispo, all the bus drivers are actually issued potassium iodide because they have the Diablo Canyon Nuclear Power Plant there, and their default is if the power plant pops there to take potassium iodide. One bus driver during an exercise decided to take a potassium iodide when they were told ahead of time, don't take it. But that's just an idea of a default protective action guide in the early phase. You have the intermediate phase. intermediate phase.

So here, source and releases been brought under control, though not necessarily. Fukushima was an interesting one because up to that point, we've always done exercises where after a day or two, the release was stopped.

But Fukushima just kept on going and going and going. It just kept on, you know, exploding with the hydrogen explosions, kept on releasing from the buildings. It was like a never-ending early phase. And that one, in my opinion, probably lasted in two weeks. It wasn't until after maybe even the first month that we really got to the intermediate phase with that. But you have to just recognize sometimes there's artificialities in your exercises and the real world kind of just surprises you. Fukushima surprised us. This can last weeks to months. And then you have the late phase. And this begins when actions can take place to reduce radiation levels in the environment to acceptable levels. I love that one because what's an acceptable level to the scientist is not necessarily the acceptable level to or to the decision maker. So that is a very interesting one where. But at least it doesn't say it will end there, but it does say it will begin where you can actually start trying to reduce radiation levels. I think it should just end there, but goes to acceptable levels. This could last for months, years. And I have another bullet. I'll put it over here. Tag levels, protection action guide levels, or dose to avoid is not appropriate here. And that's why they got rid of the 50-year protective action guide. We used to call it the first, the first second and 50-year pack. We still have the first and second, and we'll again talk about that here in the second. But know your phases. So phase one, which we call not only the early phase, but the emergency phase. So exposure pathways are pretty clear. So what in an emergency phase, the releases are going on, what are your dose pathways? Inolation, very good. Yes. Inolation, what is another one? Good, yep. So we'll say deposition. ingestion, well, ingestion, what?

Injection. Yes, yes.

Ingestion, and I'll be honest, in the early phase, ingestion is not that big of it, is not, because you're not thinking about a long-term eating at that point. But I'll put it in because that is not a bad way, but they also have in here what they call direct exposure. This could be, you know, as the plume passes through before your submersion dose, is what we sometimes called in health physics. But inhalation, deposition, and direct exposure are the typical ones that we talk about. So you have to define your exposure areas. So in this case, you want to look at the current data from environmental measurements. And you're probably going to have to make estimates of the source term using models. So this was a nuclear radiation dispersal device. What resource are you going to be using for modeling? federal resource so darn so this is early things they rack exactly so neirac is going to be doing the source modeling here they have uh depending on the emergency they have uh you know default source terms if it is from a nuclear power plant they would probably rely on nrc which would know the burn up of of the reactors and the reactor configuration very well so then then they could use that information to help estimate what the source term would be during release. And depending on the type of release. That reminds me, we've had a lot of issues with the Rascal paperwork, and I need to actually have people physically sign the documents. And I will probably email people to have come by my office to sign stuff. I meant to bring it today here. For Internet land, we'll have to figure something else out. But I wouldn't mind you guys at the very least having a copy of Rascal, it's a fun little software package now. Do you have a software you can point us down the phone? Yes, turbo for Mac. Yeah, so then there's a question. It says, are you, like, eligible to sign for licenses for your institution or whatever? If you put no, it gives you this comment and it said like, oh, then you could have someone who can't sign for licenses, then register for an account. Remind me that after class, I'll send autumn an email asking that because she said to everybody, but I thought, I thought before Yeah.

So it might be they want you they still want me to register for everybody, but then you have to register as a user. So and I think we've already I've already done that word, but thank you. So phase one tag. So principal actions are you would expect evacuation, shelter in place. There's your acronym for shelter in place, your SIP, and administration of potassium iodide.

For those of you don't remember, don't know, potassium iodide is to protect the thyroid. and this is only for a nuclear power plant release or if there is, you know for a fact, iodine is in the source term that you take potassium iodine because it's stable iodine, it will go to the thyroid and help prevent uptake of radioactive iodine into the thyroid. It's useless against everything else. And I've seen people try to say, oh, no, you should always take potassium iodide. No, only when there's a lot. iodine presence.

So it's shelter and place protective actions. So this is needed when projected doses or dose is one to five REM over four days.

So that is a protective action guide right there. One to five rim over four days, 96 hours. Sometimes people say one to five rim over 100 hours.

96 hours is close but but it's it's the same thing one to five rim over a four day period you can start and only start actions at one rim unless there has been somehow a default tag hit before that that you've already tried to evacuate to avoid that but don't anticipate that you're going to go over because that could cause its own set of problem uh usually you'll find out shelter in place is probably the best thing to do we'll talk about shielding from that as well Supplementary admin of K-I, it is a 5-REM projected. Who do you think the most sensitive population is for iodine? Infants. So dose, so protected dose to a child, probably an infant, but it just says child, thyroid from iodine exposure. Yes? I'm in place, but you need to give them tagide. How do you give it to do you? I'm not for sure. but I know that every power plan, these plans are in place. These plans are in place. As I mentioned, for instance, the bus drivers,

the public workers in San Luis Obispo are given potassium iodide so they can take it in the event that the things happen. How the public gets it as they evacuate, maybe is they go to a processing centers or something. But as I was just about to write, it is most effective if taken before the exposure. You can't take it, I think, up to two hours. After I actually have a little comment here. I'll see what the comments says. It does say right. Oh, I did actually have an old comment from find out what a child is. It was infant or toddlers. So, but I want to say the actual peg is, it's most effective clearly taken before the event. And it's still effective up to even two hours post exposure. But after that, it really doesn't give you anything because it's the point. thyroid's already done the update. Yes. Oh, this is right.

You know, first, it has iodine in it, and some people are alerted to that. So you have to be baked in order to eat it. Yes. So with potassium iodide. Yes, potassium iodide kills people. So it could probably react. Yes, yes. And so, yeah, yeah, that's why we always tell people, also, if you're going to suggest doing potassium iodide, realize that there are risk associated with that and are the risk of you getting thyroid cancer maybe 40 years in the future outweigh your risk of an allergic reaction because you yeah it takes some bit of gel and just yep cross your fingers but yeah so there are people with seafood allergies it is not recommended that they take potassium yeah yeah so evacuation versus shelter in place so evacuation is most if completed for Plume Passage. So you've made estimates that you're going to hit 1 to 5 RAM, or it could be, again, a default pag that you evacuate two miles in all directions before an event occurs, then that's most of it, which makes sense. You're avoiding any possible dose that could happen. Might increase exposure if done incorrectly. Exactly. Exactly.

So as Hayden just pointed out, if during mass evacuations, we've seen them here in Texas with hurricanes, you have traffic jams. With traffic jams, you have people stuck. And as a plume goes over or you're stuck in the middle of the plume as it goes over, you are getting dose that you wouldn't have gotten because you're probably more safe in your house or wherever you were before.

And it may be only effective protection close to a plume source. So evacuating, you know, 10, 15, 20 miles away from event, that might not be the smartest thing to do. I will point out, at least the rumor that I had heard, that during Three Mile Island, there was only one death ever associated with the Three Mile Island incident. And it was a kid who drowned. And what happened is the family relocated, evacuated, and went to their vacation home, and he drowned in the lake. had he not been evacuated or I'd have been put in that situation. But something to think about. Hold on. No, no, no, no, no, don't do that. No, no. My computer might restart. I hope. I don't want that. Turn it all. So I've got time for reboot in three minutes. Okay, we'll see what happens. So shelter and provide, okay, provides the public additional instruction. So it's not just, you know, shelter. Shelter in place, don't move. It could be turn off your air conditioning. Tape your windows shut. Or, you know, if you know that you have gaps in the wall, tape those shut. Do do something. Yeah, they'll give you more instructions about, you know, what they want you to do, sheltering in place. You know, go to the most interior room that you can, something like that. The nice thing about it, as opposed to evacuations, shelter can be implemented rapidly. I will say also the problem with shelter. shelter in place versus evacuation, depending on the type of incident, you don't need to tell people to evacuate. They will usually run for the hills. If you looked at New York following 9-11, I still remember the pictures of people crossing the bridge, Manhattan Bridge and George Washington Bridge getting away from Manhattan. And I don't think there was ever an evacuation hosted for Upper West Side, but I think a lot of people was evacuated without being told to just to get away. So evacuation may not be plausible due to weather or road conditions. And also, as I put it in big, so I'm yelling at you right now. Need to take into account an emergency responder capability if recommended is evacuation. If you do not have the resources to do a good evacuation and realize the people who are going to be doing the evacuation or manage, it are probably going to get a dose too, then shelter in place is probably your best bet. And so these are actually tables from, I think

this is actually a Fermak manual that I've taken this from, where sheltering in place or evacuation of the public, one to five rim or 10 to 50 Milleusever, projected dose over four days, evacuation or some situations sheltering in place should be initiated when projected dose is at one rim. also talks about the administration of K.I. Does it say anything about, may require, see, this one does say, KI is most effective taken prior to exposure, may require approval of state medical officials or in accordance with established emergency plan. And my computer is rebooting. But let's see if I can remember anything else about this. Hopefully there's not a lot of words after this. Okay. As my computer is rebuilt, is rebooting, I know I can at least talk about this. So what this is actually taking, this actually might be straight from EPA 400. I know this figure is also in the Fermac manual, but what it is trying to show you is there are actually shielding factors for structures that you live in. So you could actually reduce your dose of like a one story home by a factor of two to three if you just stay in your home without going out anywhere. If you have a basement, that factor is a reduction of an order of. magnitude if you have multiple and so you just should make a lot of sense because the dose is probably coming from the deposition on your roof primarily shining down on you so the further you get away from your roof is is better so that's why your basement also the ground that surrounding you is giving you good shielding if it's a three-story building the protection factor in the first story is three so it's telling you hey if you have a multi-story building go to your lowest floor to shelter in place 20, if it's a basement, three-story, notice this one, and I'd love to know why, probably because it's showing here that your dose conversion factor is best in the middle floor, and I'm betting it is because you're getting ground shine in addition to, you're getting more ground shine as you're closer in. That's what I'm betting the reason why. So you'd want a middle floor. And then you can see of course as you get larger and larger structures, even multi-story buildings, you get factors up to 100 to 200 reduction.

But my guess is it would probably, I don't know if it actually says in here, but my guess is it's going to tell you that you're not going to be having your HVAC systems on. I'm starting my computer over again.

So your take away for this slide alone should be sheltering in place does give you significant dust reduction.

I think that. there's even a dose reduction for a car. So being in a car is better than being on side. But the dose reduction is like, it's very, very small. It's nothing to write home about. But again, it's better than being in a car. Hi, everyone. I just want to continue our lecture on EPA 400 and protective action guides for radiological incident. EPA 400 is kind of important to know because it is the basis really for all protective action guides. lines that are set out in the United States following a radiological incident. And yes, the current EPA 400 does follow Fermak methodology to make calculations within their document, but the original one from 1992 does not. And so it's kind of nice that this new document is not, is at least acknowledging that EPA is not the no-all end-all for these type of incidents that they recognize that newer calculational methods and newer information might come about that will affect how protective action guides are calculated and how they might be communicated to the public. Now I'm going to continue from where we were the other day. I talked about the shielding factors and now we're going into the administration of potassium iodide. As was mentioned earlier in the previous lecture, potassium iodide is kind of a problem in that people with severe shellfish allergies can die from taking potassium iodide and therefore medical advice for at least a medical professional before taking care is really warranted because the potential of your dying from some sort of iodine exposure is probably far less than your potential from dying from some anaphylactic shock because you took K.I. and they're allergic to shellfish. So, you know, really make sure that people out there acknowledge that medical advice is definitely warranted.

So the food and drug administration for the application of potassium iodide recommends a few things.

One, they recognize children have a, are more sensitive because their cells are dividing, which are rapidly than adults, and you have to think about them probably first within these applications. So children from 0 to 18 years give K-I if a dose from radio iodines. I.

I'm just going to shorten it that with R. is 5-R-M. If you are, if you're a female and you are pregnant or lactating, give K-I if a dose is at least adults up to 40 years of age. I.

Give K. If radio nuclides, oh, sorry, radio iodine. If it says radio iodine, it is greater than or equal to 10 rim. And then over 40, 500 rim, greater than 500 rim, is the recommendation, and this is, I should say, this is a dose to the thyroid. So this is a whole body dose. If you are going to get a dose to the thyroid itself, these are the values that should not be exceeded, or it should say it should not be exceeded, that you may consider suggesting the potassium iodide might be warranted. Now, another thing that you might have to realize is how to treat emergency workers during these types of emergencies. I think I've told you this before that for an emergency worker, they must recognize or realize at least that they're probably going to get a dose during a radiological event. If they're not getting a dose, they're probably not working as hard as they probably should. And also with that dose should come some education for them, I guess prior to them getting that dose, or they must realize that even though you're getting much higher doses, than an annual, typical annual limit that it is still safe in that the risk of you getting a fatal cancer is still even well below 1% of an increase in risk from your natural incidence of getting a fatal cancer without any radiation exposure. However, during an event, you should definitely maintain ALARA. Keep your doses as low as reasonably achievable if you can.

However, there are guidelines set out there for emergencies that you should follow, and it is what we call the 5-1025 rule.

Five is the, is for all, oh sorry, I guess 5-rim, let's just change that. So 5-RM is just for occupational exposures. If you're executing normal duties, you should still get just up to 5-RM of a dose. And that is actually for an event that's actually not an annual dose. So annual doses can be actually treated completely separately. So for an event, during an event, you can get up to five, five REM. However, if you have duties that might require you to do more, such as life-saving or protection of property, your doses go up. They go up to 10-rim, protecting vital property.

What do I mean by that? Let's say like a power plant.

And if you're protecting property, you guess that you should probably protect life. And for them, prediction of life.

25 REM. Now there are new guidelines out. I would say new guidelines, but there are guidelines out there that this dose can go up to as high as 50 REM for what we consider a, I can't remember the term right now.

It's a decision dose that you can go all the way up to 50 RAM, but then you're doing your assigned documentation that you understand the risks of what is involved. And if I remember correctly, going up even to 50 REM, is still something like just increasing your risk of cancer by 4% at most. So a lot of this information, so anyways, this table kind of gives you an idea of dose limits or guidelines. It gives you an idea of what could occur if you get those types of doses. You can get more information in NCRP 165. And also during an event, you do follow. other types of guidelines as listed by the federal regulations, EPA, NRC, and DOE.

Okay, now we know, from last lecture, we know early phase packs, we know emergency worker packs, we know the PAG for potassium iodide. Now, the emergency phase is passed. We're in the intermediate phase, and now you have to think about doses to the public during this phase. And this, remember, the

intermediate phase is after, the initial emergencies over usually releases under some sort of control and you have a good idea of what's happened. So you have to worry about specific exposure pathways.

Now, for instance, whole body exposure is probably going to dominate, especially from ground shine.

So this is going to dominate during nuclear power plant accidents, radiation dispersal devices, and improvised nuclear devices.

So deposition on the ground, so you're getting the ground shine up, this is where you're going to be the most dominant doses. But you have to worry about other types of doses as well. We will worry about internal exposure.

Here, we're talking about inhalation due to resuspension. Recall resuspension is where the stuff is deposited and some sort of mechanical thing is pushing that, you know, some of the deposition back up into the air. It could be working the area, it could be wind, it could even be rain where the rain hits the ground and it causes, you know, stuff to come up into the air. Even though it's minute and it's dust, you don't see it, you're still breathing that in and you are getting a dose from the resuspendent material. You can also get internal exposure, ingestion, the material that you eat, water, food, anything that is exposed to the radiation that you take in, some amount of that material will be uptake by your body and deposited in organs and rest. released through defecation or urination. So you have to worry, though, about all of these dose pathways.

So the best ways for you to reduce dose during these things, during this phase, the intermediate phase, is one thing we call relocation, getting people out of the area, moving them out, and it couldn't be just for a couple of weeks or months until the shorter lived ready nucleides die down and they could go back in. If they wish, in most cases, they probably We won't wish even if there's just a little radiation back, you might consider doing decontamination. And this is not only for the people who are evacuating, but also the area where they want to live. Shielding.

Can shielding be used to reduce a public exposure? Time limits.

You can be outside for a given amount of time, or you could be in this area for a given amount of time. And then somehow control, spread, contamination.

This could be done by putting, you know, putting concrete over an area, putting some sort of plants that hold the, you know, hold the soil down with the root system. There's several ways that you can actually try to control the spread of surface contamination. During an exercise one time, we thought of spreading vegetable oil all over the ground, but that actually caused a problem with the EPA because they said that we were actually polluting the area more than the radiation was.

So relocation is effective, but highly disruptive, so you have to consider that when you totally move out of public. But the protective action guides for that is if somebody in the public is going to get two rim projected over the first year.

Remember, this is a person in the public. They are not following the five-rim dose limit and they are not radiation workers. So if a person in the public is going to get more than two-rim, or it's suspected they might get two-rim or more, in the first year after the event, then they should be relocated. The guidance is for them to be relocated.

The next one is 0.5 rem for every after.

We used to call this the second year pack because I think the original documentation said 500 mill rim within the second year, but really it's 500 millerum for any year following the first. So know those protective action

guides.

So the areas of relocation can be defined by environmental factors.

So, especially the plume direction. Actually, what you're seeing here is a picture called the keyhole where you might have a relocation or evacuation around the immediate area in a complete circular path, and then you might relocate individuals down the wind direction of the plume out to 5, 10, 25 miles, and that's what you're actually seeing here.

And so usually you look at relocation and you think of it in different phases. So phase one, like the emergency phase, most likely you have now evacuated these people and you don't have to worry about relocating them because they have been evacuated already.

People who were not previously evacuated, but probably now should be relocated. And then people who have been evacuated but reside outside of the relocation area and they may return home. So, you know, like I said, in the emergency phase, be evacuating for 360 degrees around but you realize now that the plume was going off let's say to the east and the west is completely clear so you can allow some public in in the northwest and south the directions of the plume outside of this keyhole here's just a listing of the intermediate phase pags uh for uh just for the public as i mentioned there's two two and 5 mill rim.

0. You actually can look at here the if you have food and when you should interdict any food if you suspect that you're going to get 0.5 rim in the in the first year whole body or five rim to any individual organ or tissue. We have drinking water limits and we also have re-entry guidelines and why you should consider re-entry following an event.

Now in the intermediate phase, you have specific priorities that you have to face. I've given you those PAG levels. Now let's think how other things might be applied and what your priorities are as a dose assessment scientist.

So first, you better determine where first year PAG or may be received.

This helps you figure out the areas that need to be evacuated.

If you know certain areas are clear, don't have radiation either do, well, definitely due to measurements in the field or to these projected models showing you that certain areas might be clear. You verify these areas are clear with field teams. So you allow previously evacuated populations to return. Allow what are called buffer zone areas to gradually return. By there is notice from the previous slide where I showed you the keyhole, right on the fringes of those areas, there might be like extra, land areas that have been evacuated because they're in specific zones, like a buffer zone. Well, you can start now shortening those buffer zones to get closer and closer to the PAG limit, if you will, at least the PAG contour line.

And again, you want to try to minimize relocation and trying to deal with a population that doesn't have a home because you've got to deal with those people. So let's see if we can do those people back in.

You begin at this moment also to evaluate dose reduction, effectiveness, simple econ techniques. What can you do to clean some of this stuff up? Very simple things. Again, this isn't the final phase, but you begin to least think about this during the intermediate phase.

You want to be able to control access to press location area.

I believe I mentioned in a previous lecture about the sensitivity of dairy cows, sometimes you must allow people back in, or people evacuated and left their pets. So they might want to go back in to get their pets, or they left some valuables that need be restructed return. So you have to think about, yes, I've

relocated, and no, you shouldn't go in there, but there are times where you have to be, you have to think reasonably, and there's always exceptions that must be considered.

You want to be able to establish monitoring and decon stations. Commit, simple, decon contaminated areas, outside relocation area. We're going to get contaminated both workers and public. How and where can you handle those decontamination? And clearly at this point, you're probably working with EPA and the National Decontamination team to figure out the best areas and best practices to do that. After all this, I guess after, and even during this, you're collecting data or collect data to establish long-term creation, rad protection criteria. Also want to be able to evaluate whether produced, like eggs, you know, you have plants and animal products, if you will, within the plume area to predicted. These are all, it's almost listed in order. I mean, really, you want to make sure people, the doses are reduced as soon as possible. So where is that two-red limit? let's set that up, let's know what we're doing. Now we can consider can people come back in the areas, either from the previously evacuated zones or the buffer zones that were evacuated or relocated. Look at what dose reduction techniques can be used to reduce doses or decon techniques can be used to reduce doses in and around the area. And you know, it all leads to finally, you know, at the very end, if now that you have food that's growing in this potentially, to have the area, is it safe to eat? Can it be, you know, eaten outside of the area? Are people going to buy it?

Probably not. If people are starving, suddenly, you know, those limits don't really seem all that of a bad thing.

So what is the final phase? It is the long-term cleanup process. At this point is no longer emergency. We are talking about EPA things right now. But phase three is the long-term process.

It's going to be used clearly for large-scale events like a nuclear power plant accident or an I and D.

Protection action guides will not be used to guide restoration or areas impacted. This is now a whole different set of guidelines which you're going to have to deal with. And then you're going to have to think about how to balance restoring businesses, putting populations back into the area.

How are you going to manage waste?

It's a completely, you know, different game at this point.

Rarely in exercises do we get to this point, we sometimes practice the changeover. Because remember, during an emergency, the Fermac is in charge of monitoring. It might not be the coordinating agency, but it is in charge of monitoring. And finally, during restoration and long-term cleanup, the Fermac switches over to the EPA. And you have to, again, very precise and dictated guidelines of when this can occur. And then this becomes the EPA's problem. And hopefully, you will not be working for these. If this was useful, again, EPA 400, very important to know, know the Pags, but as we discuss the Fermat manual coming up, these Pags are stated in there as well. So thank you very much.